A Multi-Agent Academic Helpdesk for IIT Jodhpur

Retrieval-augmented, tool-using agents on local 8B models — design and evaluation

Sreenitya Thatikunta (B23CS1072)
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Abstract

We build an academic helpdesk that answers questions about IIT Jodhpur's programmes, course contents, time table and academic rules, drafts emails to course instructors, cites the document behind every answer, and refuses — and logs for a human — anything it cannot answer. The system runs entirely on a 16 GB laptop: Google ADK agents call Ollama-hosted 8B models (Qwen3, Llama 3.1) through LiteLLM, with hybrid retrieval (dense `nomic-embed-text` + BM25) over ChromaDB and structured tools over the semester's course and exam data. The knowledge base contains 33 real documents (1,134 chunks). We compare a single-agent baseline with a multi-agent version (router + policy, schedule and email specialists) on 30 prompts (31 turns) × 2 models × 2 modes × 3 repeats. Qwen3 reaches 87% accuracy (single) and 78% groundedness (multi), Llama 3.1 65–69%; out-of-scope refusals are 100% correct and logged in all configurations. Model choice matters more than architecture; the multi-agent design improves grounding but introduces router- and specialist-specific failure modes. We also report an ADK behaviour that silently overrides specialist instructions in small models, and the data-level techniques (an amendment register for superseded rules, scoped search, deterministic guards) that mattered more than prompt wording.

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1 Problem

The assignment asks for a multi-agent *Academic Helpdesk Agent* that (i) answers questions about academic programmes, course contents, and rules and regulations, e.g. *"What is the attendance requirement to sit in the Major exam?"*, *"What is the slot for the course Autonomous Systems?"*, *"How many 400-level electives have been floated by the Humanities department this semester?"*, *"How many Open Elective credits do I need for a B.Tech?"*; (ii) drafts an email to a course instructor (sick leave, make-up permission), e.g. *"Write an email to the course instructor of CSL7570 that I need leave for 2 days"*; (iii) answers *"I don't know"* to anything else (*"What are the dining options on campus?"*) and logs such questions for a human; and (iv) cites the source document of every answer. It must use RAG with a local embedding model and vector store, tools (vector search, time-table and course lookup from CSV, instructor email lookup, email drafting), Google ADK with Ollama-hosted 8B models, run on 16 GB of RAM, start as a single agent and then become an orchestrator with policy, schedule and email agents, and be evaluated on 20–30 prompts for accuracy, groundedness and correct refusals, across models and repeated runs.

2 Data

2.1 Documents

We collected 33 real documents (Table 1). 24 are public pages and PDFs from `iitj.ac.in`; 9 come from the author's institute mailbox (Registration and Examination Cell circulars) because the current time table, course list and IH lists are not public. Three scanned circulars were OCR'd with Tesseract. Every document is described in `data/manifest.csv` with a title, category, source, date and a *status*: 29 are *current*, 4 *superseded* by newer versions (2018 B.Tech regulations, 2025–26 calendar, 2023 slot chart, the previous year's IH list).

Category	#	Examples
Regulations	5	Academic Regulations 2019 (PDF + web), B.Tech Regulations 2018, PG Regulations 2022+, MS by Research
Circulars / orders	5	Circular 51 (attendance, 2024), Circular 52 (exam policy, 2024), M.Tech norms, announcements
Curricula	5	B.Tech CSE (pre-2026 batches), AIDE 2026+, EE 2025+, M.Tech AI, M.Tech CSE
Course catalogues	8	CSE course details, HSS concept note, common courses, IH2/IH3 syllabi, IH course lists
Calendars	3	AY 2026–27, AY 2025–26, Summer 2026
Time tables	3	Finalized course list, slot system + classroom allocation + minor-exam grid, 2023 slot chart
Exams	2	Major-exam time table Sem I 2026–27, New Guidelines for Institute Examination (2024)
Registration	2	UG registration guidelines, Circular 2283

Table 1: The 33 source documents by category.

2.2 Structured data

The semester spreadsheets are converted to CSVs: `courses.csv` (821 offerings from the final classroom allocation plus the finalized course list, with slot, classroom, instructor, level and an `ih_group` tag so

that "Humanities" means *Institute Humanities*, including IH courses taught by AIDE), `slots.csv` (slot → day and time), `exam_schedule.csv` (45 minor + 44 major exam slot entries) and `faculty.csv` (312 instructors). **All email addresses are dummy values** (`first.last@iitj.example`, a reserved domain); other addresses in the documents are masked at ingestion, and sheets listing other students were not collected.

2.3 Amendment register

Institutional rules are amended by later circulars while the old text stays in the regulations. `data/amendments.csv` lists passages superseded by a named newer document (e.g. the 2019 attendance rule → Circular 51; the 2019/2022 make-up rules → the 2024 Examination Guidelines); matching chunks are marked *amended* and never retrieved.

3 System design

3.1 Retrieval pipeline

Documents are loaded page by page (PDF, OCR text, HTML with tables rendered as rows, spreadsheets per sheet), chunked to ∼1,800 characters with 250 overlap (documents under 3,200 characters stay whole so a circular's "existing" and "revised" policy are not separated), prefixed with a contextual header `[title | date | page]`, embedded with `nomic-embed-text` and stored in ChromaDB. A query runs dense (top 30) and BM25 (top 30) retrieval, fused with reciprocal rank fusion; amended chunks are dropped, superseded documents get half weight, at most two chunks per document are returned, and if the best dense distance exceeds 0.35 the search reports *not found*. A `document` filter restricts the search to one document (e.g. a programme's curriculum).

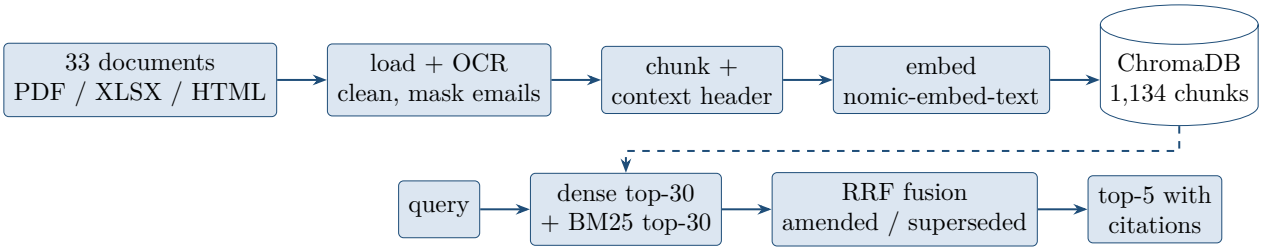


Figure 1: Ingestion (top) and hybrid retrieval (bottom); retrieval reads the ChromaDB index (dashed).

Retrieval was tuned on a separate 19-question check: dense-only search placed the right document first for 9/18 questions (17/18 in the top 5); BM25 alone made it worse (15/18) until contextual headers gave keywords something to match; the final pipeline reaches 12/19 at rank 1 and 18/19 in the top 5. Best-match distances of in-scope (0.15–0.28) and out-of-scope questions (0.26–0.43) overlap, so the distance threshold rejects only half of the out-of-scope questions — refusals must come from the agents.

3.2 Tools

Tool	Behaviour
`search_documents`	hybrid RAG search; passages with citation, date, status
`lookup_course`	course by code or name; unknown codes return "did you mean" suggestions (CSL7570 → CSL7650)
`get_slot`	slot, weekly class times, classroom, instructors, minor/major exam date
`list_courses`	count and list of courses floated by department / level ("Humanities" = IH)
`get_instructor_email`	instructor and (dummy) email by course or name
`draft_email`	saves the draft to `drafts/` with a NOT SENT header; never sends
`log_unanswered_question`	appends the question to `logs/unanswered.jsonl` for a human

Table 2: Agent tools (plain Python functions; ADK derives the schemas from type hints and docstrings).

3.3 Agents

The **single agent** is one ADK `LlmAgent` with all seven tools and a prompt composed of blocks: scope, *call a tool before every answer*, grounding rules (exact refusal sentence + log; a related passage is not an answer; the newest source and the revised policy win; cite 1–2 sources), and sections for policy, schedule and email questions. The **multi-agent** version (Figure 2) has a router that calls `transfer_to_agent` or refuses, and three specialists that see only their own tools and rules. Models run with temperature 0.2 and an 8,192-token context; Qwen3 runs with thinking disabled (`think=False`), which removed leaked reasoning from its answers and cut latency tenfold.

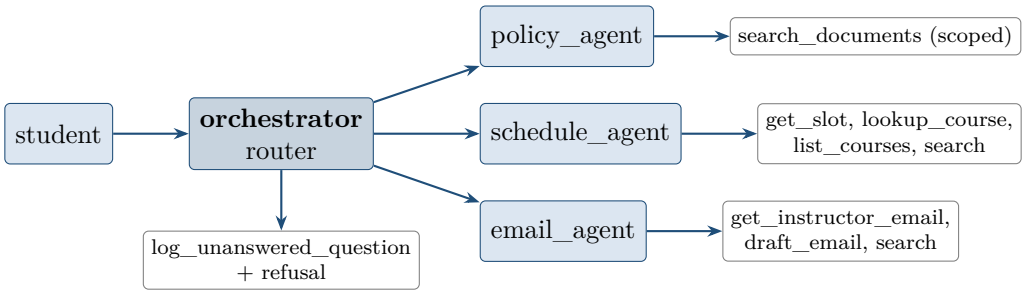


Figure 2: Multi-agent design. Only the router transfers; every new student message returns to it.

3.4 Guards

Two deterministic callbacks wrap every agent. (1) If a model writes the refusal sentence without calling `log_unanswered_question` — Qwen3 did so in about half of its refusals during development — the question is logged in code, so every unanswered question reaches a human. (2) Tool calls that Llama 3.1 prints as JSON text are stripped from the answer.

4 Engineering findings during development

Prompt rules are not enough for 8B models. The first single-agent run answered the attendance and Open Elective questions *without calling any tool*, with invented citations. A *tools-first* block fixed tool use, but grounded answers were still wrong when old and new versions of a rule were both retrieved: three of five passages for the attendance question stated the 2019 rule. The amendment register, keeping short circulars whole and the per-document cap fixed this where prompt wording ("the newest source wins") had not.

Parameters a small model can misuse should be removed. A `course_type` filter turned "3 Humanities electives" into 0 (most departments leave the category blank), and the `document` filter

was used unprompted to scope rule questions to the wrong document. Both were removed from the agents that misused them.

ADK's transfer instructions override specialist rules. In multi-agent mode the policy agent never asked for the student's programme (0/3), although its prompt alone did (3/3). The LLM request showed that, when an agent is allowed to transfer, ADK appends generic instructions to the *end* of its system prompt (*"If you are the best to answer the question according to your description, you can answer it"*) and adds a `transfer_to_agent` tool. Removing the transfer bookkeeping messages did not help; disabling transfers from specialists did (3/3, then 5/5). Only the router routes, and multi-turn conversations still work because each new message is routed again.

Programme- and batch-dependent answers need clarification. Open Elective credits are 10 in the 2019 regulations and the CSE curriculum but 12 in the newer EE and AIDE curricula. The agent therefore asks for the branch and year of joining and then searches only that curriculum ("B.Tech CSE, joined 2023" → 10 credits).

5 Evaluation methodology

The evaluation set has 30 items (31 turns, Appendix A): policy 9, schedule 5, course counts 3, email 4, out-of-scope 6, multi-intent 2, and one two-turn conversation; it contains all five assignment examples. Every expected answer was checked against the documents or tools. Each turn specifies an expected *behaviour* (answer, refuse, draft, suggest, clarify), regular expressions for the required facts, forbidden patterns for superseded rules, acceptable citation sources, the relevant tools, and the expected specialist. Each of the 12 configurations (2 models × 2 modes × 3 repeats) ran all items in fresh sessions with its own draft folder and unanswered log (372 turns, no errors).

- **Accuracy**: the expected behaviour is met (facts present and no superseded rule; refusal; a clarifying question without a credit number; a suggestion without drafting; a saved draft to the right recipient).
- **Groundedness** (answer and draft turns): correct, the relevant tool was called, and an expected source is cited (for drafts: the recipient came from the instructor lookup).
- **Refusals**: correct-refusal rate on out-of-scope items, false-refusal rate on the rest, logging rate.
- **Routing**: the first specialist the router transfers to is the expected one.
- **Consistency**: whether correctness, or the facts the question asks for, differ across repeats.

Every turn scored wrong or ungrounded was read by hand. This found three errors in the answer key, corrected for all configurations: the PhD maximum duration *is* stated in the PG Regulations (§11.5, 14 registered semesters; the item had been designed as unanswerable after a keyword search for "years" missed it), and two over-literal patterns ("two exams: Minor and Major"; a quoted "I" grade). No per-answer overrides were needed.

6 Results

Config	Accuracy	Grounded	Correct ref.	False ref.	Routing	Latency mean / p90 (s)
single / qwen3	87% (81/93)	70% (48/69)	100% (18/18)	4% (3/75)	–	9.5 / 19.9
multi / qwen3	84% (78/93)	78% (54/69)	100% (18/18)	8% (6/75)	94% (87/93)	12.2 / 31.3
single / llama3.1	65% (60/93)	46% (32/69)	100% (18/18)	1% (1/75)	–	14.2 / 27.6
multi / llama3.1	69% (64/93)	62% (43/69)	100% (18/18)	0% (0/75)	94% (87/93)	13.3 / 27.1

Table 3: Headline metrics, all repeats pooled (93 turns per configuration; 69 answer/draft turns for groundedness; 18 out-of-scope turns). All refused out-of-scope questions were logged.

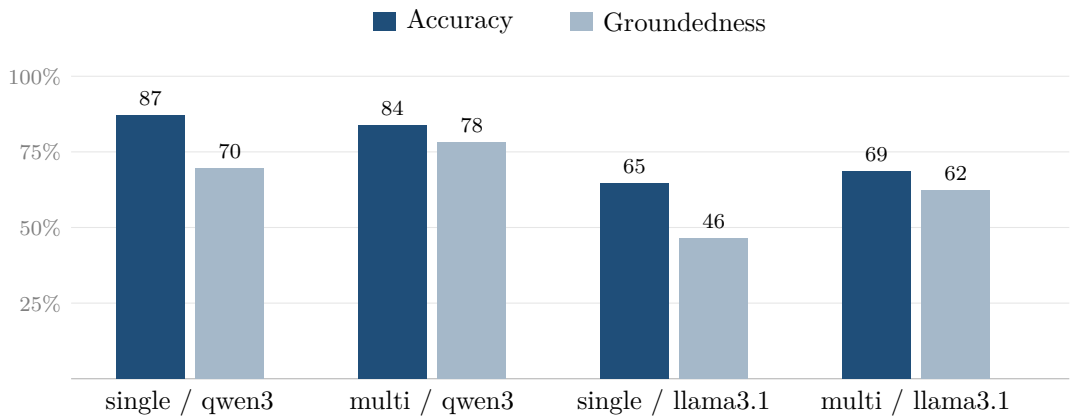


Figure 3: Accuracy and groundedness per configuration.

Config	Policy	Schedule	Count	Email	Out of scope	Multi-intent	Conversation
single / qwen3	78	100	67	100	100	50	100
multi / qwen3	78	80	67	100	100	50	100
single / llama3.1	52	100	67	25	100	17	50
multi / llama3.1	70	100	67	25	100	0	50

Table 4: Accuracy (%) by category.

Questions whose answers changed across runs

Config	Item	Correct (r1 r2 r3)	Asked-for facts per repeat
single / llama3.1	M02	✗✗✓	`22 | nov | - ‖ 22 | nov | - ‖ 22 | nov | 75%`
single / llama3.1	P01	✗✗✗	`75% | - | STALE ‖ 75% | - | STALE ‖ 75% | -`
single / llama3.1	P06	✓✗✓	`minor | major ‖ minor | major | STALE ‖ minor | major`
single / llama3.1	P07	✗✗✗	`- | 20 | - ‖ - | 20 | - ‖ REFUSED`
single / llama3.1	P08	✓✗✗	`navigation ‖ - ‖ -`
single / llama3.1	P09	✓✗✓	`14 ‖ - ‖ 14`
single / llama3.1	V01.1	✓✗✓	`ASKED ‖ NOT_ASKED ‖ ASKED`
single / llama3.1	V01.2	✗✓✗	`- | STALE ‖ 10 ‖ -`
multi / llama3.1	M02	✗✗✗	`- | - | 75% ‖ - | - | 75% ‖ 22 | - | 75%`
multi / llama3.1	P01	✗✗✗	`75% | - | STALE ‖ 75% | - ‖ 75% | - | STALE`
multi / llama3.1	P04	✓✗✗	`i"grade ‖ - ‖ -`

Qwen3 gave the same asked-for facts in every repeat in both modes; all changes are Llama 3.1's (e.g. V01: asking for the programme in two repeats but not the third; P09: "14 semesters" twice and an invented "4–5 years" once).

Routing errors (multi-agent)

Config	Item	Routed to	Repeats
multi / llama3.1	M01	policy_agent	3/3
multi / llama3.1	P03	email_agent	3/3
multi / qwen3	P09	helpdesk_orchestrator	3/3
multi / qwen3	S02	helpdesk_orchestrator	3/3

Table 7: All routing errors are systematic (the same item in every repeat). For Qwen3 P09 and S02 the router handled the request itself — a refusal and an empty reply respectively.

Config	Unstable correctness	Answer changed
single / qwen3	0/31	0/31
multi / qwen3	0/31	0/31
single / llama3.1	6/31	8/31
multi / llama3.1	1/31	3/31

Table 5: Consistency across the three repeats (31 turns). *Answer changed*: the facts the question asks for differ between repeats.

7 Analysis

Model choice matters more than architecture. Qwen3 (thinking disabled) beats Llama 3.1 by 15–22 accuracy points in both modes, is faster in single mode, and is fully stable across repeats. Llama 3.1 never called `draft_email` in any of its 24 draft requests — it asked again for details the student had already given or wrote the email into the chat — and answered the attendance question with the *existing* policy quoted inside Circular 51 in all six runs.

Multi-agent improves grounding, not necessarily accuracy. Focused specialists search and cite more consistently (groundedness +8 points for Qwen3, +16 for Llama 3.1), and for Llama 3.1 accuracy (+4) and stability (answers changed on 3 instead of 8 questions) also improve. For Qwen3 accuracy drops by 3 points through three failure modes that do not exist in the single agent: the router returning an empty reply (S02), a specialist refusing a syllabus question whose answer it had retrieved (P08), and the router refusing an in-scope question itself (P09). The extra routing step adds about 2.7 s on average for Qwen3.

Refusals. Out-of-scope questions are refused and logged in every configuration (72/72); the deterministic guard makes the logging independent of the model. The remaining problem is *false* refusals of answerable questions (8% for multi-agent Qwen3).

Groundedness vs. accuracy. Several answers were right but ungrounded: Qwen3 answered the I-grade question without any tool call and cited a non-existent "Academic Regulations 2026–27, Section 4.3"; the attendance half of a mixed request was answered from the model's own knowledge. Reporting groundedness separately exposes answers that are right by luck.

Hardest items. The IH2 course count failed everywhere (0/12): models read "5th semester" as 500-level courses, and `list_courses` cannot filter by IH group. The make-up rule + email request failed everywhere (0/12): the single agent skipped the draft, and the multi-agent Qwen3 drafted correctly but omitted the 10-day deadline.

8 Limitations and future work

- Add IH-group and semester filters to `list_courses`; add a router fallback when the router returns no text and no transfer; soften the "related passage is not an answer" rule for syllabus questions.
- The distance threshold separates out-of-scope queries poorly; an LLM relevance check on the top passage or a small classifier could replace it.
- Regex scoring needed three answer-key corrections; an LLM judge with human spot checks would scale better.
- Time tables are tentative and semester-specific; the (idempotent) data pipeline must be re-run each semester.

- Only two 8B models were compared; larger or tool-tuned models are the obvious next step.

9 Conclusion

A fully local helpdesk on 8B models can answer most academic questions correctly with citations, refuse and log everything outside its scope, and draft instructor emails safely. With Qwen3 it reaches 87% accuracy (single agent) and 78% groundedness (multi-agent) with stable answers across runs. The improvements that mattered most were in the data and tools — explicit recency through an amendment register, scoped search, removing misusable parameters, deterministic guards — and in understanding what the agent framework adds to the prompt. The multi-agent architecture improves grounding but needs its own safeguards against router and specialist failures.

Code and data: the repository README describes setup and reproduction; `docs/TECHNICAL.md` gives the technical details and `docs/PROGRESS_LOG.md` the full development log.

A Evaluation questions

Item	Category	Question	Expected
P01	policy	What is the attendance requirement to sit in the Major exam?	*answer*: At least 75% attendance in each course (full attendance expected); below 75% without valid reason the instructor can recommend de-registration (Circular 51, 2024, supersedes the old rule).
P02	policy	What happens if I am absent from all classes for four weeks in a semester?	*answer*: Absence from all academic activities for 4+ weeks (with or without leave) results in automatic semester withdrawal / cancellation of registration.
P03	policy	I missed a minor exam because I was sick. How do I apply for a make-up exam?	*answer*: Make-up is for a missed Minor; submit the Makeup/I-grade application (Annexure A) with a medical certificate issued/verified by the IITJ Medical Officer to the Office of Academic Affairs within 10 days.
P04	policy	What should I do if I miss the Major exam due to illness?	*answer*: Apply for an I grade (not a make-up); the I grade application must reach the Office of Academic Affairs before the last day of the examinations, with a verified medical certificate.
P05	policy	What is the maximum duration allowed to complete a B.Tech programme?	*answer*: 14 registered semesters (2019 Academic Regulations §11.4; supersedes the older 8-year rule).
P06	policy	How many examinations are held for a course in a semester?	*answer*: One Minor and one Major examination (Circular 52, Jul 2024; Examination Guidelines 2024).
P07	policy	When are the minor examinations for Semester I of 2026-27?	*answer*: 15-20 September 2026 (tentative).
P08	policy	What topics are covered in the Autonomous Systems course?	*answer*: Autonomous/semi-autonomous systems (mobile robots, cars): e.g. navigation, obstacle avoidance, planning, multi-agent systems, game theory.
P09	policy	What is the maximum duration allowed to complete a PhD?	*answer*: 14 registered semesters, which may include up to 6 semesters of Registration kept Alive (RKA) (PG Regulations 2022 onwards, §11.5).

Item	Category	Question	Expected
S01	schedule	What is the slot for the course Autonomous Systems?	*answer*: CSL7650, slot E (Tue/Thu/Fri 11:00-11:50), LHC 106.
S02	schedule	Who teaches Health Economics this semester, and in which classroom?	*answer*: LAL4070, Alok Ranjan, LHC 204 (slot D).
S03	schedule	When is the major exam for CSL7650?	*answer*: 22 November 2026 (Sunday), 09:00 AM-12:00 PM, tentative.
S04	schedule	In which slot and classroom is the course Digital Society held?	*answer*: LAL3080, slot N, LHC 105 (Rachel Philip).
S05	schedule	What is the slot for CSL9999?	*answer*: No such course is offered this semester (no suggestions).
C01	count	How many 400 level electives have been floated by the Humanities department in this semester?	*answer*: 3: LAL4070 Health Economics, LAL4850 Introduction to Climate Change Economics, AIL4070 Fundamentals of Economics (IH3).
C02	count	How many IH2 courses are offered to 5th semester students this semester?	*answer*: 6: LAL3030, LAL3060, LAL3080, LAL3090, LAL3040, LAL3130.
C03	count	How many 700-level courses is the CSE department offering this semester?	*answer*: 16.
E01	email	Write an email to the course instructor of CSL7570 that I need leave for 2 days	*suggest*: CSL7570 is not offered; 'Did you mean CSL7650 (Autonomous Systems)?' - no draft
E02	email	Write an email to the instructor of Autonomous Systems saying I need sick leave for 3 days.	*draft*: Draft to gaurav.harit@iitj.example (sick leave, 3 days), saved, not sent
E03	email	Draft an email to the Health Economics instructor asking permission for a make-up exam, as I missed the minor exam due to fever.	*draft*: Draft to alok.ranjan@iitj.example (make-up for missed minor), saved, not sent
E04	email	Write an email to Prof. Rachel Philip saying I will miss tomorrow's Digital Society class due to a family emergency.	*draft*: Draft to rachel.philip@iitj.example, saved, not sent
O01	out of scope	What are the available dining options on the campus?	*refuse*: I don't know + logged
O02	out of scope	What is the best way to reach the campus from Jodhpur city center?	*refuse*: I don't know + logged
O03	out of scope	What will the weather be like in Jodhpur tomorrow?	*refuse*: I don't know + logged
O04	out of scope	How do I apply for a hostel room change?	*refuse*: I don't know + logged
O05	out of scope	Who won the cricket World Cup in 2023?	*refuse*: I don't know + logged
O06	out of scope	What is the average placement package for CSE students?	*refuse*: I don't know + logged
M01	multi intent	I missed my minor exam of CSL7650 because I was ill. What is the rule for a make-up exam? Also write an email to the instructor asking for a make-up exam.	*draft*: Rule: make-up for missed Minor, apply within 10 days with verified medical certificate (2024 Guidelines) + draft to gaurav.harit@iitj.example
M02	multi intent	When is the major exam for Autonomous Systems, and what is the attendance requirement to sit in it?	*answer*: 22 November 2026, 09:00-12:00 (tentative); attendance >= 75%, else deregistration may be recommended (Circular 51).
V01.1	conversation	How many Open Elective credits do I need to earn in order to get a B.Tech. Degree?	*clarify*: Asks for B.Tech branch and year of joining (credits differ by programme/batch)

Item	Category	Question	Expected
V01.2	conversation	I am in B.Tech CSE, joined in 2023.	*answer*: 10 OE credits (B.Tech CSE curriculum, batches before 2026, p. 10).

B Per-question results

Correct runs out of 3 repeats (**bold**: all correct; red: none correct).

Item	single / qwen3	multi / qwen3	single / llama3.1	multi / llama3.1
C01	**3/3**	**3/3**	**3/3**	**3/3**
C02	0/3	0/3	0/3	0/3
C03	**3/3**	**3/3**	**3/3**	**3/3**
E01	**3/3**	**3/3**	**3/3**	**3/3**
E02	**3/3**	**3/3**	0/3	0/3
E03	**3/3**	**3/3**	0/3	0/3
E04	**3/3**	**3/3**	0/3	0/3
M01	0/3	0/3	0/3	0/3
M02	**3/3**	**3/3**	1/3	0/3
O01	**3/3**	**3/3**	**3/3**	**3/3**
O02	**3/3**	**3/3**	**3/3**	**3/3**
O03	**3/3**	**3/3**	**3/3**	**3/3**
O04	**3/3**	**3/3**	**3/3**	**3/3**
O05	**3/3**	**3/3**	**3/3**	**3/3**
O06	**3/3**	**3/3**	**3/3**	**3/3**
P01	**3/3**	**3/3**	0/3	0/3
P02	**3/3**	**3/3**	**3/3**	**3/3**
P03	0/3	**3/3**	0/3	0/3
P04	**3/3**	**3/3**	**3/3**	1/3
P05	**3/3**	**3/3**	**3/3**	**3/3**
P06	**3/3**	**3/3**	2/3	**3/3**
P07	**3/3**	**3/3**	0/3	**3/3**
P08	**3/3**	0/3	1/3	**3/3**
P09	0/3	0/3	2/3	**3/3**
S01	**3/3**	**3/3**	**3/3**	**3/3**
S02	**3/3**	0/3	**3/3**	**3/3**
S03	**3/3**	**3/3**	**3/3**	**3/3**
S04	**3/3**	**3/3**	**3/3**	**3/3**
S05	**3/3**	**3/3**	**3/3**	**3/3**
V01.1	**3/3**	**3/3**	2/3	0/3
V01.2	**3/3**	**3/3**	1/3	**3/3**